

Lesson 01 After-Lesson Practice

Square and Cube Roots: practice and self-test

Name: _____

Date: _____

Before you start

Estimated time: 20–30 minutes. **Calculator:** only where a question says *check*.

Required: Retrieval R1, Easy Q1–Q5, Medium Q6–Q10 and the Self-Test.

Optional challenge: Hard Q11–Q13.

Short on time? Do R1, Q1, Q2, Q5, Q6, Q8 and the Self-Test.

Stuck? Use the method from class: (1) square or cube numbers either side, (2) which is it closer to?, (3) trap it with decimals, (4) choose the closer one and check.

Retrieval

R1. Work out: (a) 2^4 (b) $(-5)^2$ (c) 0.3^2

Easy: build confidence

Q1. Find: (a) $\sqrt{36}$ (b) $\sqrt{144}$ (c) $\sqrt{169}$

Q2. Find: (a) $\sqrt[3]{27}$ (b) $\sqrt[3]{125}$ (c) $\sqrt[3]{1}$

Q3. A square has an area of 100 cm^2 . How long is one side?

Q4. A cube has a volume of 64 cm^3 . How long is one edge?

Q5. Between which two consecutive whole numbers does each root lie?

(a) $\sqrt{12}$ (b) $\sqrt{75}$ (c) $\sqrt[3]{40}$

Medium: show understanding

- Q6.** Estimate $\sqrt{30}$ to 1 decimal place. Show the square numbers you used and your decimal check. Then check with a calculator.

- Q7.** Estimate $\sqrt[3]{70}$ to 1 decimal place. Show your working. Then check with a calculator.

- Q8.** Which is larger, $\sqrt{50}$ or $\sqrt[3]{300}$? Explain *without* a calculator.

- Q9.** You know $\sqrt{25} = 5$ and $\sqrt[3]{8} = 2$. Use these facts to write down:

(a) $\sqrt{2500}$ (b) $\sqrt{0.25}$ (c) $\sqrt[3]{8000}$

- Q10.** (a) Find $\sqrt[3]{-64}$. (b) Find $\sqrt[3]{-1}$.
(c) Explain why no real number is equal to $\sqrt{-16}$. *Hint: what do 4^2 and $(-4)^2$ equal?*

Hard: apply and reason (optional if time is short)

- Q11.** A square rug has an area of 12 m^2 . It must fit along a wall that is 3.5 m long. Will the rug fit? Justify your answer *without* a calculator.

- Q12.** Leo writes: $\sqrt{36 + 64} = \sqrt{36} + \sqrt{64} = 6 + 8 = 14$.
Find Leo's first error, explain it, and write the correct answer.

- Q13.** A cuboid measures $4 \text{ cm} \times 4 \text{ cm} \times 8 \text{ cm}$. A cube has the same volume as the cuboid. Estimate the edge length of the cube to 1 decimal place.

Self-Test (no calculator)

Self-Test: three mixed questions

ST1. Find $\sqrt{121}$.

ST2. Between which two consecutive whole numbers is $\sqrt[3]{90}$?

ST3. Estimate $\sqrt{18}$ to 1 decimal place. Show a check.

Reflection

I can do independently ...	I need to practise ...	My question is ...